

Euler–Run Spectrum for Fixed-Lag Centered Möbius Capacity

RH research program

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Abstract

Fix a lag $h \geq 1$. A centered phase table reads $\mu_0(n-h), \mu(n), \mu(n+h)$, and universal safety forbids two positive outputs at separation $2h$ for every ternary word. The complete three-shift terminal-log law converts every fixed-table limit, for every terminal clock, into collision-aware squarefree densities. Positive projection and relation saturation then give an exact tropical trace on all eight subsets of $\{-1, 0, +1\}$ for every finite clock. The familiar four-state compression is proved throughout the non-self-loop regime $q \nmid 2h$; when $q \mid 2h$, the full eight states remain necessary in general.

On square-support clocks, a phasewise identity for each shared ternary value charges two consecutive transitions by one center weight. Once the base support contains $p_0(h) = \min\{p \text{ prime} : p \nmid 2h\}$, the centered capacity equals a raw step- $2h$ path-MWIS normalized by the squarefree density. Exact bracketed-run densities yield the Euler–run endpoint

$$B_\infty(h) = \frac{3}{\pi^2} + \frac{1}{2} \sum_{\substack{1 \leq \ell < p_0(h)^2 \\ \ell \text{ odd}}} R_{\ell, h}.$$

For every fixed h , this is the supremum over finite clocks, and no finite clock attains it. Fresh-prime lifts can plateau: strict gain is equivalent to the presence of an even positive run, while a CRT construction guarantees eventual strict gain. Across fixed lags, $\inf_{h \geq 1} B_\infty(h) = 3/\pi^2$, and the infimum is not attained. No monotonicity, supremum, or maximum in h is claimed.

Keywords: Möbius function; terminal logarithmic average; centered local rule; fixed lag; tropical trace; squarefree run; nonattained supremum.

1 Definitions and main results

Write $\mu_0(k) = \mu(k)$ for integers $k \geq 1$, and $\mu_0(k) = 0$ for $k \leq 0$. A *terminal clock* is a function

$$1 \leq \omega(X) \leq X, \quad \omega(X) \longrightarrow \infty. \quad (1)$$

Fix $h, q \geq 1$, put $d = 2h$, and fix, before $X \rightarrow \infty$, one sign table for every phase,

$$F_r : \{-1, 0, +1\}^3 \longrightarrow \{-1, +1\}, \quad r \in \mathbb{Z}/q\mathbb{Z}.$$

Its centered output and scored terminal functional are

$$\varepsilon_F(n) = F_{n \bmod q}(\mu_0(n-h), \mu(n), \mu(n+h)), \quad (2)$$

$$L_{h, q, X}(F) = \frac{1}{\log \omega(X)} \sum_{X/\omega(X) < n \leq X} \frac{\mu(n) \varepsilon_F(n)}{n}. \quad (3)$$

Thus $\varepsilon_F(n)$ is the output, while $\mu(n) \varepsilon_F(n)$ is the score. The family is *universally distance- d safe* when

$$\neg(F_r(a, b, c) = +1 \text{ and } F_{r+d}(c, e, f) = +1) \quad (4)$$

for every phase r and every $a, b, c, e, f \in \{-1, 0, +1\}$. The shared letter is $c = \mu(n + h) = \mu((n + d) - h)$.

The density notation must retain collisions among the three shifts. Let $\mathcal{I} = \{L, C, R\}$ and

$$a_L = h, \quad a_C = 0, \quad a_R = -h. \quad (5)$$

For $S \subseteq \mathcal{I}$, first deduplicate modulo p^2 , then set

$$B_{p,S} = \{a_i \bmod p^2 : i \in S\}, \quad \nu_{p,S} = |B_{p,S}|, \quad \tau_{p,S}(r) = \#\{b \in B_{p,S} : b \equiv r \pmod{p}\}. \quad (6)$$

Define

$$\Theta_{h,q,r}(S) = \frac{1}{q} \prod_{p \nmid q} \left(1 - \frac{\nu_{p,S}}{p^2}\right) \prod_{p \parallel q} \left(1 - \frac{\tau_{p,S}(r)}{p}\right) \prod_{p^2 \mid q} \mathbf{1}_{\{r \bmod p^2 \notin B_{p,S}\}}, \quad (7)$$

$$\kappa_h(S) = \prod_p \left(1 - \frac{\nu_{p,S}}{p^2}\right). \quad (8)$$

Then $\Theta_{h,q,r}(\emptyset) = 1/q$ and

$$\sum_{r \bmod q} \Theta_{h,q,r}(S) = \kappa_h(S). \quad (9)$$

Only in a collision-free j -site case does this reduce to

$$K_j = \prod_p \left(1 - \frac{j}{p^2}\right), \quad K_0 = 1, \quad K_1 = \frac{6}{\pi^2}. \quad (10)$$

The phase-resolved exact-support density is

$$\Pi_{h,q,r}(U) = \sum_{W \subseteq \mathcal{I} \setminus U} (-1)^{|W|} \Theta_{h,q,r}(U \cup W). \quad (11)$$

It is nonnegative and has mass $\sum_{U \subseteq \mathcal{I}} \Pi_{h,q,r}(U) = 1/q$.

For $x, y \in \{-1, 0, +1\}$, write

$$S(x, y) = \{C\} \cup (\{L\} : x \neq 0) \cup (\{R\} : y \neq 0) \quad (12)$$

and define the nonnegative cell weight and transition

$$\lambda_{h,q,r}(x, y) = 2^{-\mathbf{1}_{\{x \neq 0\}} - \mathbf{1}_{\{y \neq 0\}}} \Pi_{h,q,r}(S(x, y)), \quad (13)$$

$$\mathcal{K}_{h,q,r}(U, V) = \sum_{\substack{x \notin U \\ y \in V}} \lambda_{h,q,r}(x, y), \quad U, V \subseteq \{-1, 0, +1\}. \quad (14)$$

Theorem 1.1 (Fixed-clock terminal law and full tropical formula). *For every fixed h, q, F , the limit $L_{h,q}(F) = \lim_{X \rightarrow \infty} L_{h,q,X}(F)$ exists for every terminal clock (1) and has the same value for all such clocks. Form the finite maximum only after these fixed-table limits:*

$$C_h(q) = \max_{F \text{ universally safe}} |L_{h,q}(F)|. \quad (15)$$

If γ runs over the $\gcd(q, d)$ cycles of $r \mapsto r + d$ in $\mathbb{Z}/q\mathbb{Z}$, then

$$C_h(q) = \sum_{\gamma} \max_{Y_r \subseteq \{-1, 0, +1\}} \sum_{r \in \gamma} \mathcal{K}_{h,q,r}(Y_{r-d}, Y_r). \quad (16)$$

Thus the exact formula is a tropical trace on eight states for every finite q . Both signs of every nonzero optimum are attained.

Theorem 1.2 (Four-state scope and self-loop obstruction). *If $q \nmid d$, an optimizer in (16) can be chosen from the four antipodally symmetric subsets of $\{-1, 0, +1\}$. If $q \mid d$, every cycle is a self-loop and the theorem retains all eight states; no four-state reduction is asserted. This exception is strict in general: for $h = 2, q = 4$, the full and four-state-restricted coefficient vectors in the basis (K_0, K_1, K_2, K_3) are respectively*

$$(0, 0, \tfrac{1}{2}, -\tfrac{1}{2}) \quad \text{and} \quad (0, 0, 1, -2), \quad (17)$$

and the full value is strictly larger.

For the endpoint, let

$$p_0(h) = \min\{p \text{ prime} : p \nmid d\}. \quad (18)$$

For a finite set $J \subset \mathbb{Z}$, define the absolutely convergent local product

$$D_h(J) = \prod_p \left(1 - \frac{|\{dj \bmod p^2 : j \in J\}|}{p^2} \right), \quad (19)$$

and, for $\ell \geq 1$,

$$\begin{aligned} R_{\ell,h} &= D_h([0, \ell - 1]) - D_h(\{-1\} \cup [0, \ell - 1]) \\ &\quad - D_h([0, \ell]) + D_h([-1, \ell]). \end{aligned} \quad (20)$$

Intervals here denote sets of integers. The quantity $R_{\ell,h}$ is the density of an exact bracketed squarefree run of length ℓ in step- d coordinates, hence is nonnegative. Set

$$B_\infty(h) = \frac{3}{\pi^2} + \frac{1}{2} \sum_{\substack{1 \leq \ell < p_0(h)^2 \\ \ell \text{ odd}}} R_{\ell,h}. \quad (21)$$

Theorem 1.3 (Euler–run endpoint and finite nonattainment). *For every fixed $h \geq 1$,*

$$\boxed{\sup_{q < \infty} C_h(q) = B_\infty(h), \quad C_h(q) < B_\infty(h) \quad (q < \infty).} \quad (22)$$

The supremum is over finite clocks only and is taken after the fixed-table terminal limits and finite safe maxima.

Corollary 1.4 (Lag infimum). *Across fixed positive integer lags,*

$$\boxed{\inf_{h \geq 1} B_\infty(h) = \frac{3}{\pi^2}.} \quad (23)$$

Every fixed h satisfies $B_\infty(h) > 3/\pi^2$, so the infimum is not attained. No supremum, maximum, or monotonicity assertion in h is made.

2 Terminal table bridge and relation saturation

RH-394 proves the complete terminal-log table law for any three fixed distinct shifts: Theorem 1.1, equation (8), Theorem 1.2, equations (11)–(14), and Corollary 1.3 (printed/PDF pages 2–3) [2]. We instantiate it at the ordered shifts $(a_L, a_C, a_R) = (h, 0, -h)$. They are distinct because $h \geq 1$, and all of h, q, F are fixed before the limit. This is the sole analytic terminal-clock input in the present paper.

To state the resulting bridge, put

$$H_r(x, z, y) = zF_r(x, z, y)$$

and, for $U \subseteq \mathcal{I}$, let

$$\overline{H}_{r,U} = 2^{-|U|} \sum_{\sigma \in \{-1,+1\}^U} H_r(\sigma_U, 0_{U^c}).$$

The cited law, including the boundary convention μ_0 , gives

$$L_{h,q}(F) = \sum_{r \bmod q} \sum_{U \subseteq \mathcal{I}} \Pi_{h,q,r}(U) \overline{H}_{r,U}. \quad (24)$$

It also gives (7)–(9). The deduplication in (6) is essential: if, for example, $p^2 \mid d$, several coordinate shifts can define the same forbidden class modulo p^2 . Equation (24) is qualitative and fixed-data; no rate uniform in h or q is imported. Tao and Tao–Teräväinen are inherited two-point and odd-correlation provenance inside RH-394, not separately strengthened inputs here [4, 5].

Lemma 2.1 (Positive projection). *Given a safe family F , replace every $+1$ output at a cell whose center z is not $+1$ by -1 , leaving every other cell unchanged. Call the result F^+ . Then F^+ is safe and*

$$L_{h,q}(F^+) \geq L_{h,q}(F).$$

If

$$A_r = \{(x, y) \in \{-1, 0, +1\}^2 : F_r^+(x, +1, y) = +1\}, \quad (25)$$

then

$$L_{h,q}(F^+) = \sum_{r \bmod q} \sum_{(x,y) \in A_r} \lambda_{h,q,r}(x, y). \quad (26)$$

Proof. At $z = -1$, changing the output from $+1$ to -1 changes the score zF from -1 to $+1$; at $z = 0$, it does not change the score. Deleting positive outputs cannot create a forbidden pair, so safety is preserved. This establishes the pointwise score comparison before any limit is taken.

Start now from the all-minus table. Its score is $-z$, whose uniform average on every exact-support sign stratum is zero. Turning the center- $+1$ cell $(x, +1, y)$ from minus to plus changes H by two. The density of that specified sign cell is

$$2^{-1-\mathbf{1}_{\{x \neq 0\}}-\mathbf{1}_{\{y \neq 0\}}} \Pi_{h,q,r}(S(x, y)).$$

Multiplication by two gives exactly (13). Summing the changed cells proves (26). $\square$

For a relation $A \subseteq \{-1, 0, +1\}^2$, define, before using either symbol,

$$\text{Source}(A) = \{x : (x, y) \in A \text{ for some } y\}, \quad \text{Target}(A) = \{y : (x, y) \in A \text{ for some } x\}.$$

Lemma 2.2 (Safety and saturation). *The projected relation family is safe if and only if*

$$\text{Target}(A_r) \cap \text{Source}(A_{r+d}) = \emptyset \quad (r \bmod q). \quad (27)$$

Put $Y_r = \text{Target}(A_r)$. Then

$$A_r \subseteq (\{-1, 0, +1\} \setminus Y_{r-d}) \times Y_r. \quad (28)$$

Replacing every A_r by the saturated relation

$$\tilde{A}_r = (\{-1, 0, +1\} \setminus Y_{r-d}) \times Y_r \quad (29)$$

preserves safety and weakly increases (26). Conversely, every subset profile $(Y_r)_{r \bmod q}$ in (29) is safe.

Proof. An edge $(x, c) \in A_r$ and an edge $(c, y) \in A_{r+d}$ are precisely two positive outputs allowed by a common shared letter c . Their absence for every $c \in \{-1, 0, +1\}$ is (27). It follows that every source of A_r lies outside $\text{Target}(A_{r-d}) = Y_{r-d}$, while its targets lie in Y_r , proving (28). All weights in (26) are nonnegative, so saturation cannot lower the score. Finally, an edge of $\tilde{A}_r$ ends in Y_r , while an edge of $\tilde{A}_{r+d}$ begins outside Y_r ; hence the saturated family is safe. $\square$

Lemma 2.3 (Reflection). *Define*

$$F_r^\rho(x, z, y) = F_r(-x, -z, -y).$$

This map preserves universal safety and satisfies

$$L_{h,q}(F^\rho) = -L_{h,q}(F). \quad (30)$$

Proof. Simultaneous negation is a bijection of the ternary words appearing in (4), so it preserves safety. Moreover,

$$zF_r^\rho(x, z, y) = -H_r(-x, -z, -y).$$

Every exact-support stratum in (24) is invariant under simultaneous sign negation. Its uniform average therefore changes sign, which proves (30). This is a terminal table-law identity; it does not identify the reflected table with a pointwise transformation of the observed Möbius sequence. $\square$

3 The tropical optimizer and four-state boundary

Proposition 3.1 (Exact eight-state trace). *The right side of (16) is the maximum positive terminal score over all safe tables.*

Proof. By Lemmas 2.1 and 2.2, a positive optimum is attained by a saturated subset profile. The phase r contribution is

$$\sum_{\substack{x \notin Y_{r-d} \\ y \in Y_r}} \lambda_{h,q,r}(x, y) = \mathcal{K}_{h,q,r}(Y_{r-d}, Y_r).$$

Addition by d decomposes $\mathbb{Z}/q\mathbb{Z}$ into $\gcd(q, d)$ cycles, each of length $q/\gcd(q, d)$. Profiles on distinct cycles do not interact, so their maxima add. Cyclic closure of each finite dynamic program is exactly the tropical trace in (16). There are eight possible subsets of $\{-1, 0, +1\}$, including all singleton states and all self-loops. $\square$

Proof of Theorem 1.1. The existence and clock independence of every fixed-table limit follow from (24). Positive projection, saturation, and Proposition 3.1 give the displayed positive maximum. Lemma 2.3 shows that every score has a safe score of the opposite sign, hence the positive maximum equals the absolute capacity and both orientations are attained. This proves every assertion of the theorem, including self-loop clocks. $\square$

For an antipodally symmetric set $Y \subseteq \{-1, 0, +1\}$, write

$$u(Y) = (u_0(Y), u_1(Y)) = \left(\mathbf{1}_{\{0 \in Y\}}, \frac{|Y \cap \{-1, +1\}|}{2} \right) \in \{0, 1\}^2.$$

Abbreviate

$$\alpha_r = \Pi_{h,q,r}(\{C\}), \quad \beta_r = \Pi_{h,q,r}(\{L, C\}), \quad \gamma_r = \Pi_{h,q,r}(\{C, R\}), \quad \eta_r = \Pi_{h,q,r}(\{L, C, R\}).$$

Directly summing (13) gives, on the four states,

$$\begin{aligned} \mathcal{K}_{h,q,r}(U, V) &= \alpha_r(1 - u_0(U))u_0(V) + \beta_r(1 - u_1(U))u_0(V) \\ &\quad + \gamma_r(1 - u_0(U))u_1(V) + \eta_r(1 - u_1(U))u_1(V). \end{aligned} \quad (31)$$

Proposition 3.2 (Non-self-loop compression). *If $q \nmid d$, the eight-state maximum has a four-state optimizer, and its transitions are given by (31).*

Proof. Fix the membership of zero in every Y_r , and set $k_r = |Y_r \cap \{-1, +1\}| \in \{0, 1, 2\}$. The cell weights depend on a nonzero outer coordinate only through its being nonzero, not through its sign. Consequently the two transition terms in which Y_r occurs are affine in k_r once all other memberships are fixed.

The hypothesis $q \nmid d$ is equivalent to $q/\gcd(q, d) > 1$. Thus the occurrence of Y_r as target in the r -term and as excluded source in the $r + d$ -term belongs to two distinct terms, with no self-identification. An affine function at $k_r = 1$ is no larger than at least one endpoint $k_r = 0$ or 2. Round one phase at a time. The objective never decreases, and after finitely many rounds all nonzero memberships are antipodally symmetric. Equation (31) follows by splitting the sum into the four zero/nonzero support patterns. $\square$

Proof of Theorem 1.2. Proposition 3.2 proves the non-self-loop statement. If $q \mid d$, then $r + d = r$ for every phase. A self-loop transition uses the same nonzero count both as an included target and an excluded source; terms proportional to $k_r(2 - k_r)$ can occur, so the preceding affine rounding is unavailable.

The obstruction $(h, q) = (2, 4)$ can be checked without approximation. At each even phase every self-loop transition is zero: phase 0 has zero center mass, while at phase 2 both outer coordinates are forced to be zero and cannot lie simultaneously outside and inside the same state. At either odd phase, substituting (7)–(14) for all eight subsets gives

Y type	number of states	self-loop value
$\emptyset, \{-1, 0, +1\}$	2	0
one nonzero sign, or its complement with 0	4	$(K_2 - K_3)/4$
$\{0\}$ or $\{-1, +1\}$	2	$K_2/2 - K_3$

Here

$$\frac{K_3}{K_2} = \frac{1}{2} \prod_{p \geq 3} \left(1 - \frac{1}{p^2 - 2}\right).$$

The product is less than one. On the other hand,

$$\sum_{p \geq 3} \frac{1}{p^2 - 2} < \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{1}{n(n-1)} = 1 - \log 2 < \frac{1}{3},$$

and $\prod(1 - x_i) \geq 1 - \sum x_i$, together with $\log 2 > 2/3$, gives

$$\frac{1}{3} < \frac{K_3}{K_2} < \frac{1}{2}. \quad (32)$$

Thus a one-sign state is the full optimum at each odd phase, whereas the four-state optimum is $\{0\}$ or $\{-1, +1\}$. Adding the two odd phases yields respectively $(K_2 - K_3)/2$ and $K_2 - 2K_3$, which are the coefficient vectors in (17); their difference $(3K_3 - K_2)/2$ is positive by (32). $\square$

4 Square-support marginal charge

Let $\mathcal{P}$ be a nonempty finite set of primes and

$$q_{\mathcal{P}} = \prod_{p \in \mathcal{P}} p^2.$$

More generally, call Q *square-supported* if every prime divisor of Q occurs to exponent at least two. Its positive center phases form

$$V_h(Q) = \{r \bmod Q : \Theta_{h,Q,r}(\{C\}) > 0\} = \{r \bmod Q : p^2 \nmid r \text{ for every } p \mid Q\}.$$

Write

$$N_h(Q) = |V_h(Q)|, \quad (33)$$

$$\alpha_h(Q) = \max_{\substack{I \subseteq V_h(Q) \\ I \cap (I+d) = \emptyset}} |I|, \quad (34)$$

$$M_h(Q) = K_1 \frac{\alpha_h(Q)}{N_h(Q)}. \quad (35)$$

The normalization is deliberate: $\alpha_h(Q)$ is a raw, unweighted MWIS cardinality, whereas $M_h(Q)$ is a weighted terminal value. Neither (35) nor the equality with centered capacity below is being asserted for clocks having a prime only to the first power.

Lemma 4.1 (Clock divisibility and one-site embedding). *If $q \mid Q$, then $C_h(q) \leq C_h(Q)$. On every square-supported Q , one also has $M_h(Q) \leq C_h(Q)$.*

Proof. Regard a q -phase table as a Q -phase table by literal repetition. It produces the same output word, preserves universal safety, and has the same fixed-table terminal limit. The declared clock need not be a minimal period. Maximizing after the limits gives the first assertion.

For the second, choose an independent set I in (34). At $r \in I$, let the projected relation be all of $\{-1, 0, +1\}^2$; off I , let it be empty. This family is safe, and summing (13) over both outer coordinates gives $\Theta_{h,Q,r}(\{C\})$. All positive center phases of a square-supported clock have the common weight

$$\delta_Q = \frac{1}{Q} \prod_{p \mid Q} \left(1 - \frac{1}{p^2}\right) = \frac{K_1}{N_h(Q)}. \quad (36)$$

Thus the embedded value is $\delta_Q \alpha_h(Q) = M_h(Q)$. $\square$

Lemma 4.2 (Per-state shared marginal). *Let Q be square-supported, and suppose that both r and $r + d$ belong to $V_h(Q)$. For each $t \in \{-1, 0, +1\}$, define*

$$m_r(t) = \sum_{x \in \{-1, 0, +1\}} \lambda_{h,Q,r}(x, t), \quad m'_{r+d}(t) = \sum_{y \in \{-1, 0, +1\}} \lambda_{h,Q,r+d}(t, y).$$

Then

$$\begin{aligned} m_r(0) &= \Theta_{h,Q,r}(\{C\}) - \Theta_{h,Q,r}(\{C, R\}) \\ &= \Theta_{h,Q,r+d}(\{C\}) - \Theta_{h,Q,r+d}(\{L, C\}) = m'_{r+d}(0), \end{aligned} \quad (37)$$

$$m_r(\pm 1) = \frac{1}{2} \Theta_{h,Q,r}(\{C, R\}) = \frac{1}{2} \Theta_{h,Q,r+d}(\{L, C\}) = m'_{r+d}(\pm 1). \quad (38)$$

In particular,

$$m_r(t) = m'_{r+d}(t) \quad (t \in \{-1, 0, +1\}), \quad \sum_{t \in \{-1, 0, +1\}} m_r(t) = \delta_Q. \quad (39)$$

Proof. Summing exact-support cells with right coordinate zero gives the event that the center is squarefree while the right site is not; this is the difference in the first line of (37). If the right coordinate is a specified nonzero sign, uniform sign splitting contributes the factor 1/2 in (38). The same argument at phase $r + d$ uses the left coordinate.

It remains to prove the two phase identities. Outside the prime support of Q , the pair of coordinate shifts $\{C, R\} = \{0, -h\}$ is a translate of $\{L, C\} = \{h, 0\}$, so the deduplicated local

factors are identical even when the two residues collide. At a supported prime square, the first pair requires $r \not\equiv 0, -h \pmod{p^2}$, and the second requires $r + d \not\equiv 0, h \pmod{p^2}$. Since the endpoint center phases r and $r + d$ are positive, in both cases the only remaining condition is $r + h \not\equiv 0 \pmod{p^2}$. Hence

$$\Theta_{h,Q,r}(\{C, R\}) = \Theta_{h,Q,r+d}(\{L, C\}).$$

Both singleton center densities equal δ_Q by (36). This proves (37) and (38); adding the three states proves (39). The $t = 0$ difference is part of the identity, not an omitted zero-weight case. $\square$

Lemma 4.3 (Pair and path charge). *Under the hypotheses of Lemma 4.2, for all $U, V, W \subseteq \{-1, 0, +1\}$,*

$$\mathcal{K}_{h,Q,r}(U, V) + \mathcal{K}_{h,Q,r+d}(V, W) \leq \delta_Q. \quad (40)$$

Consequently a positive step- d path of length L contributes at most $\lceil L/2 \rceil \delta_Q$ to (16).

Proof. Dropping the outer-coordinate restrictions in the two transitions and splitting at the shared state gives

$$\begin{aligned} \mathcal{K}_{h,Q,r}(U, V) &\leq \sum_{t \in V} m_r(t), \\ \mathcal{K}_{h,Q,r+d}(V, W) &\leq \sum_{t \notin V} m'_{r+d}(t). \end{aligned}$$

Their sum is δ_Q by (39). Pair successive sites of a path, leaving one unpaired site when L is odd; a single transition is at most its full center mass δ_Q . This gives the path charge. $\square$

Proposition 4.4 (Square-supported centered saturation). *If Q is square-supported and $p_0(h) \mid Q$, then*

$$C_h(Q) = M_h(Q) = K_1 \frac{\alpha_h(Q)}{N_h(Q)}. \quad (41)$$

Proof. Because $p_0 \nmid d$, a step- d orbit modulo p_0^2 visits every residue class. The factor $p_0^2 \mid Q$ therefore forces at least one zero center phase on every step- d cycle. Deleting the zero phases splits the positive graph into finite paths; there are no all-positive cycles. Lemma 4.3 bounds the contribution of each path by its ordinary path-MWIS cardinality times δ_Q . Summing paths gives

$$C_h(Q) \leq \delta_Q \alpha_h(Q) = M_h(Q).$$

The reverse inequality is the one-site embedding from Lemma 4.1. $\square$

Proposition 4.5 (Same-support cover). *Suppose $p_0(h) \in \mathcal{P}$, $q_{\mathcal{P}} \mid Q$, and $\text{rad}(Q) = \text{rad}(q_{\mathcal{P}})$. Put $R = Q/q_{\mathcal{P}}$. No coprimality condition on R and d is required. Then*

$$N_h(Q) = R N_h(q_{\mathcal{P}}), \quad \alpha_h(Q) = R \alpha_h(q_{\mathcal{P}}), \quad (42)$$

$$C_h(Q) = M_h(Q) = M_h(q_{\mathcal{P}}) = C_h(q_{\mathcal{P}}). \quad (43)$$

Proof. Reduction modulo $q_{\mathcal{P}}$ makes the positive support word on $\mathbb{Z}/Q\mathbb{Z}$ an R -fold cover of the corresponding word on $\mathbb{Z}/q_{\mathcal{P}}\mathbb{Z}$, proving the first count in (42). By Proposition 4.4, the base positive graph is a disjoint union of paths. The inverse image of a path under any graph cover is R disjoint copies of that path: choose a lift of one endpoint, then lift its unique successive edges. This argument is independent of $\gcd(R, d)$. Hence the path-MWIS cardinality scales by R , proving the second count. The ratio in (35) is unchanged, and Proposition 4.4 at both clocks proves (43). $\square$

Remark 4.6 (Why the p_0 qualification is necessary). For $h = 6$, one has $d = 12$ and $p_0 = 5$. The same-support lift $36 \mid 72$ occurs before the support contains 5^2 :

$$(\alpha_6(36), N_6(36)) = (9, 24), \quad (\alpha_6(72), N_6(72)) = (24, 48).$$

The raw ratios are $3/8$ and $1/2$, so unconditional same-support scaling is false. Once the base includes p_0^2 ,

$$(\alpha_6(900), N_6(900)) = (291, 576), \quad (\alpha_6(1800), N_6(1800)) = (582, 1152),$$

in agreement with Proposition 4.5. These exact counts are finite illustrations of the proved path-cover mechanism, not asymptotic evidence.

The finite MWIS and square-clock viewpoint follows the combinatorial template of RH-375, Theorem 2.2 and Sections 3–5 [1]. The $h = 1$ centered marginal argument in RH-395 is the direct finite precedent [3]. Neither predecessor supplies the general distance- d , collision-aware proof above, and neither is used as a terminal-clock analytic input.

5 Euler–run formula

For a finite prime set $\mathcal{P}$ and finite $J \subset \mathbb{Z}$, define

$$D_{h,\mathcal{P}}(J) = \prod_{p \in \mathcal{P}} \left(1 - \frac{|\{dj \bmod p^2 : j \in J\}|}{p^2} \right), \quad (44)$$

and define $R_{\ell,h,\mathcal{P}}$ by the same four-term expression as (20), with D_h replaced by $D_{h,\mathcal{P}}$. Thus

$$\begin{aligned} R_{\ell,h,\mathcal{P}} &= D_{h,\mathcal{P}}([0, \ell - 1]) - D_{h,\mathcal{P}}(\{-1\} \cup [0, \ell - 1]) \\ &\quad - D_{h,\mathcal{P}}([0, \ell]) + D_{h,\mathcal{P}}([-1, \ell]). \end{aligned} \quad (45)$$

Lemma 5.1 (Exact finite run densities). *Suppose $p_0(h) \in \mathcal{P}$. On $\mathbb{Z}/q_{\mathcal{P}}\mathbb{Z}$, the exact number of bracketed positive step- d runs of length ℓ is*

$$q_{\mathcal{P}} R_{\ell,h,\mathcal{P}}. \quad (46)$$

These quantities are nonnegative integers and vanish for $\ell \geq p_0(h)^2$. Moreover,

$$\alpha_h(q_{\mathcal{P}}) = \frac{N_h(q_{\mathcal{P}})}{2} + \frac{q_{\mathcal{P}}}{2} \sum_{\substack{1 \leq \ell < p_0(h)^2 \\ \ell \text{ odd}}} R_{\ell,h,\mathcal{P}}. \quad (47)$$

Proof. For $r \bmod q_{\mathcal{P}}$, all phases $r + dj$, $j \in J$, are positive precisely when, for each $p \in \mathcal{P}$, the residue $r \pmod{p^2}$ avoids the distinct classes $-dj \pmod{p^2}$. The Chinese remainder theorem therefore gives the exact density (44). Inclusion–exclusion on the two endpoint events $r - d$ positive and $r + d\ell$ positive gives (45). Its complement interpretation is exactly: zero at index -1 , positive at indices $0, \dots, \ell - 1$, and zero at index ℓ . This proves (46) and nonnegativity.

Because $p_0 \nmid d$, the residues $dj \pmod{p_0^2}$ for any p_0^2 consecutive indices j cover all classes. Every such block therefore contains a phase divisible by p_0^2 , proving the run cutoff. There are no all-positive cycles.

Finally, a path of length ℓ has MWIS cardinality

$$\left\lceil \frac{\ell}{2} \right\rceil = \frac{\ell}{2} + \frac{1}{2} \mathbf{1}_{\{\ell \text{ odd}\}}.$$

Summing the first term over all runs gives $N_h/2$, and summing the second counts the odd runs. Equation (46) now gives (47). $\square$

Proposition 5.2 (Finite Euler–run endpoint). *If $p_0(h) \in \mathcal{P}$, then*

$$C_h(q\mathcal{P}) = M_h(q\mathcal{P}) = B_{h,\mathcal{P}}, \quad (48)$$

$$B_{h,\mathcal{P}} = \frac{K_1}{2} + \frac{K_1}{2D_{h,\mathcal{P}}(\{0\})} \sum_{\substack{1 \leq \ell < p_0(h)^2 \\ \ell \text{ odd}}} R_{\ell,h,\mathcal{P}}. \quad (49)$$

Proof. The singleton finite density satisfies

$$D_{h,\mathcal{P}}(\{0\}) = \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2}\right) = \frac{N_h(q\mathcal{P})}{q\mathcal{P}}.$$

Insert (47) into the normalized definition (35). This gives (49). Proposition 4.4 gives (48). $\square$

Lemma 5.3 (Infinite densities and termwise limit). *For every fixed finite $J \subset \mathbb{Z}$, the products $D_{h,\mathcal{P}}(J)$ converge to $D_h(J)$ as $\mathcal{P}$ increases through the primes. The limit is the density of simultaneous squarefreeness at the step- d positions indexed by J . Consequently*

$$R_{\ell,h,\mathcal{P}} \longrightarrow R_{\ell,h} \geq 0 \quad (\ell \text{ fixed}).$$

Proof. The number of excluded classes at a prime is at most $|J|$, and

$$\sum_p \frac{|\{dj \bmod p^2 : j \in J\}|}{p^2} \leq |J| \sum_p \frac{1}{p^2} < \infty.$$

Thus the local products converge absolutely in the standard product sense (with a possible zero caused only by a finite local factor). Finite CRT gives every truncated density. Removing the truncation is justified by the square-divisor union bound

$$\bar{d}\{n : p^2 \mid n + dj \text{ for some } j \in J, p > Y\} \leq |J| \sum_{p > Y} \frac{1}{p^2} \longrightarrow 0.$$

Hence the product is the claimed density. Applying the same two-endpoint inclusion–exclusion as in Lemma 5.1 shows that the four-term limit is an exact bracketed-run density, so it is nonnegative. $\square$

Proposition 5.4 (Cofinal convergence). *Let $\mathcal{P}_y$ be any increasing prime-initial family containing $p_0(h)$ and tending to all primes. Then*

$$B_{h,\mathcal{P}_y} \longrightarrow B_\infty(h). \quad (50)$$

Proof. Lemma 5.3 gives

$$D_{h,\mathcal{P}_y}(\{0\}) \longrightarrow D_h(\{0\}) = \prod_p (1 - p^{-2}) = K_1.$$

It also gives convergence of every run-density term. The run cutoff $\ell < p_0(h)^2$ is fixed and finite, so the limit can be taken termwise in (49). Its first term is $K_1/2 = 3/\pi^2$, and its run coefficient tends to $1/2$. The result is exactly (21). $\square$

Remark 5.5 (Certified numerical orientation only). The exact rational interval oracle in the companion artifact encloses

$$B_\infty(1) \in [0.4214, 0.4224], \quad B_\infty(2) \in [0.3282, 0.3296], \quad B_\infty(3) \in [0.4143, 0.4181].$$

These intervals orient the scale of the endpoint. They are not used in any comparison or proof, and they imply no monotonicity in h .

6 Fresh-prime lifts and strict nonattainment

Fix a finite prime support $\mathcal{P}$ containing $p_0(h)$, and let $\varpi \notin \mathcal{P}$ be a prime with $\varpi \nmid dq_{\mathcal{P}}$. Put $\mathcal{P}' = \mathcal{P} \cup \{\varpi\}$. Every old positive run has length less than p_0^2 , and

$$p_0^2 < \varpi^2,$$

because p_0 is already supported and no smaller prime can be both fresh and coprime to d .

Lemma 6.1 (Deletion parity on one lifted path). *Let P_L be a path with vertices $1, \dots, L$. Deleting vertex j leaves MWIS cardinality*

$$\left\lceil \frac{j-1}{2} \right\rceil + \left\lceil \frac{L-j}{2} \right\rceil. \quad (51)$$

If L is odd, this is one less than $\lceil L/2 \rceil$ exactly when j is odd. If L is even, it always equals $L/2$.

Proof. Deleting j splits P_L into paths of lengths $j-1$ and $L-j$; their MWIS cardinalities add, giving (51). Substitution of $L = 2m+1$ or $L = 2m$ gives the two parity statements. $\square$

Proposition 6.2 (Fresh-prime recurrence). *Let $\mathcal{R}_{\mathcal{P}}$ be the multiset of old positive run lengths and set*

$$E_{\mathcal{P}} = \sum_{\substack{L \in \mathcal{R}_{\mathcal{P}} \\ L \text{ even}}} \frac{L}{2}, \quad O_{\mathcal{P}} = \sum_{\substack{L \in \mathcal{R}_{\mathcal{P}} \\ L \text{ odd}}} \left\lceil \frac{L}{2} \right\rceil.$$

Then

$$N_h(q_{\mathcal{P}'}) = (\varpi^2 - 1)N_h(q_{\mathcal{P}}), \quad (52)$$

$$\begin{aligned} \alpha_h(q_{\mathcal{P}'}) &= \varpi^2 \alpha_h(q_{\mathcal{P}}) - O_{\mathcal{P}} \\ &= (\varpi^2 - 1)\alpha_h(q_{\mathcal{P}}) + E_{\mathcal{P}}, \end{aligned} \quad (53)$$

$$B_{h,\mathcal{P}'} - B_{h,\mathcal{P}} = \frac{K_1 E_{\mathcal{P}}}{(\varpi^2 - 1)N_h(q_{\mathcal{P}})}. \quad (54)$$

Thus a fresh-prime step is strict exactly when the old positive graph has an even run. It is not necessarily strict at every step.

Proof. Because $\varpi \nmid q_{\mathcal{P}}$, every old phase has ϖ^2 lifts modulo $q_{\mathcal{P}'}$. Exactly one lift is divisible by ϖ^2 , proving (52).

Consider an old run of length $L < \varpi^2$. Among its ϖ^2 lifted copies, $\varpi^2 - L$ have no deletion, and for each $j = 1, \dots, L$ exactly one copy has its j th vertex deleted. Indeed, the lift parameter runs bijectively modulo ϖ^2 ; two different positions cannot be deleted in the same lift because that would give $\varpi^2 \mid d(j_1 - j_2)$, while $\varpi \nmid d$ and $0 < |j_1 - j_2| < \varpi^2$.

Lemma 6.1 shows that an odd run loses one unit for each odd deletion position, a total loss $\lceil L/2 \rceil$, whereas an even run loses nothing. Its total lifted contribution is therefore

$$\begin{cases} (\varpi^2 - 1)\lceil L/2 \rceil, & L \text{ odd,} \\ \varpi^2 L/2, & L \text{ even.} \end{cases}$$

Summing runs gives both forms of (53). Dividing (53) by (52) in (35) proves (54). $\square$

Remark 6.3 (A genuine plateau). For $h = 9$, exact raw counts give

$$(\alpha_9(36), N_9(36)) = (16, 24), \quad (\alpha_9(900), N_9(900)) = (384, 576),$$

and therefore

$$M_9(36) = M_9(900) = \frac{2K_1}{3}.$$

The first clock does not yet contain $p_0(9)^2 = 25$, so this fixture is outside the recurrence's qualified base domain. It nevertheless rules out any blanket claim of strict gain at every square-support prime step. Proposition 6.2, rather than stepwise strictness, is the exact statement.

Lemma 6.4 (CRT creation of an even run). *Every finite support containing $p_0(h)$ has a finite extension whose positive graph contains an exact run of length two.*

Proof. Enlarge the support, if necessary, until it contains two distinct primes a, b with $a \nmid d$ and $b \nmid d$. Seek a starting phase r whose interior phases $r, r + d$ are positive and whose endpoints $r - d, r + 2d$ are zero. Impose

$$r \equiv d \pmod{a^2}, \quad r \equiv -2d \pmod{b^2}.$$

The left endpoint is then divisible by a^2 , and the right endpoint is divisible by b^2 . Since neither prime divides d , the two interior phases are nonzero modulo both prime squares. At every other supported prime p , choose a local residue avoiding the two classes $0, -d \pmod{p^2}$; at least two of the p^2 classes remain even for $p = 2$. The Chinese remainder theorem combines these choices. The resulting r begins an exact positive run of length two. $\square$

Proposition 6.5 (Eventual strictness). *For every finite $\mathcal{P}$ containing $p_0(h)$, some finite extension $\mathcal{P}^*$ satisfies*

$$B_{h, \mathcal{P}^*} > B_{h, \mathcal{P}}.$$

Consequently $B_{h, \mathcal{P}} < B_\infty(h)$ whenever $\mathcal{P}$ also contains every prime divisor of d .

Proof. Add generic primes until Lemma 6.4 applies. Every such addition is nondecreasing by (54). The extended graph now has an even run, so adjoining one further fresh prime is strict, again by (54). This proves the first assertion.

For the second, extend $\mathcal{P}$ through the missing primes in increasing order. Since all prime divisors of d were already included, every newly added prime is coprime to d , so the recurrence applies at every step. The values are nondecreasing, some future step is strict, and Proposition 5.4 identifies their limit as $B_\infty(h)$. Hence the starting value is strictly below that limit. $\square$

Proof of Theorem 1.3. First let $\mathcal{P}_y$ be a cofinal prime-initial family containing $p_0(h)$. Proposition 5.2 and (50) give

$$C_h(q_{\mathcal{P}_y}) = B_{h, \mathcal{P}_y} \longrightarrow B_\infty(h).$$

Therefore $\sup_q C_h(q) \geq B_\infty(h)$.

For the strict upper bound, fix an arbitrary finite q . Choose a finite prime set $\mathcal{P}$ containing $p_0(h)$, every prime divisor of q , and every prime divisor of d . Set

$$Q = \text{lcm}(q, q_{\mathcal{P}}).$$

Then $q \mid Q$, $q_{\mathcal{P}} \mid Q$, and $\text{rad}(Q) = \text{rad}(q_{\mathcal{P}})$. Lemma 4.1, Proposition 4.5, and Proposition 6.5 give

$$C_h(q) \leq C_h(Q) = M_h(q_{\mathcal{P}}) = B_{h, \mathcal{P}} < B_\infty(h).$$

This proves strict nonattainment at every finite clock and the reverse supremum inequality. At no point is q allowed to depend on X ; the scalar supremum is taken only after all fixed-table limits and finite maxima. $\square$

7 The landscape across fixed lags

Lemma 7.1 (An isolated run has positive density). *For every fixed $h \geq 1$,*

$$R_{1,h} > 0.$$

Proof. Choose distinct primes a, b with $a \nmid d$ and $b \nmid d$. The Chinese remainder theorem gives one residue class modulo $a^2 b^2$ satisfying

$$r \equiv d \pmod{a^2}, \quad r \equiv -d \pmod{b^2}. \quad (55)$$

Every integer in this class has $a^2 \mid r - d$ and $b^2 \mid r + d$. Moreover, its center is nonzero modulo both a^2 and b^2 , because neither prime divides d . Among these integers, impose for each remaining prime p the local avoidance $r \not\equiv 0 \pmod{p^2}$. Finite CRT counts followed by the elementary square-divisor tail give this subevent the density

$$\frac{1}{a^2 b^2} \prod_{p \neq a, b} \left(1 - \frac{1}{p^2}\right) > 0.$$

Here the product is positive because $\sum_p p^{-2} < \infty$. On the subevent, the center r is squarefree while the two step- d neighbors $r - d$ and $r + d$ are not. It is therefore contained in the exact bracketed length-one event whose density is $R_{1,h}$. $\square$

Lemma 7.2 (Outside-prime boundary bound). *For every integer $Y \geq 2$, put*

$$d_Y = \prod_{\substack{p \leq Y \\ p \text{ prime}}} p^2, \quad h_Y = \frac{d_Y}{2}. \quad (56)$$

Then h_Y is a positive integer and

$$0 \leq B_\infty(h_Y) - \frac{3}{\pi^2} \leq \frac{1}{2} \sum_{p > Y} \frac{1}{p^2} < \frac{1}{2(Y-1)}. \quad (57)$$

Proof. The factor 2^2 occurs in d_Y , so h_Y is integral and $2h_Y = d_Y$. Exact positive runs in step- d_Y coordinates partition the run-start event

$$\{r : r \text{ is squarefree and } r - d_Y \text{ is not squarefree}\}.$$

All runs are finite: the first prime not dividing d_Y supplies the run-length cutoff proved above. Consequently the density of this event is

$$\sum_{\ell \geq 1} R_{\ell, h_Y}. \quad (58)$$

If $p \leq Y$ and $p^2 \mid r - d_Y$, then $p^2 \mid d_Y$ forces $p^2 \mid r$, contrary to the center being squarefree. Every boundary in (58) must therefore be supplied by a prime $p > Y$. The union bound and the density p^{-2} of one square divisibility class give

$$\sum_{\ell \geq 1} R_{\ell, h_Y} \leq \sum_{p > Y} \frac{1}{p^2}.$$

The endpoint bonus uses only odd runs, so nonnegativity of all the R_{ℓ, h_Y} and (21) prove the first inequality in (57). Finally, $\sum_{p > Y} p^{-2} \leq \sum_{n > Y} n^{-2} < 1/(Y-1)$, which proves the displayed strict bound. $\square$

Proof of Corollary 1.4. All run densities in (21) are nonnegative, so $B_\infty(h) \geq 3/\pi^2$. Lemma 7.1 puts the strictly positive term $R_{1,h}/2$ into the endpoint for every fixed h , because $1 < p_0(h)^2$. Hence equality is impossible at any fixed lag.

On the other hand, Lemma 7.2 supplies a sequence of individually fixed integer lags h_Y for which the endpoint bonus tends to zero. Thus values approach $3/\pi^2$ from above, proving (23) and nonattainment. This argument compares scalar endpoints only after their fixed-lag limits; it provides neither a varying-lag terminal limit nor an ordering of endpoints by lag. $\square$

8 Source roles, reproducibility, and limitations

8.1 Analytic and combinatorial provenance

The sole analytic input is the complete three-shift terminal-log law of RH-394, frozen at repository commit `6b3d616851cd2d7cba66371d0aa9f25b8e8bf2f7` [2]. Section 2 uses precisely its fixed-shift density, phase-average, mass, and terminal-clock conclusions. The present paper supplies every subsequent safety, tropical, square-clock, run-density, strictness, and lag-landscape argument.

RH-395, frozen at commit `20de7202518f4488cbd9c7d63bf94aaa3dc94476`, is cited only as the $h = 1$ finite relation-saturation and tropical-optimizer precedent [3]. RH-375, frozen at commit `071fed1b2a5d8488b9d2e35a99a753953b233584`, is cited only as the finite one-site MWIS and square-clock template [1]. Neither provides an analytic terminal-clock input here. The Tao two-point and Tao–Teräväinen odd-correlation papers are inherited provenance through RH-394 [4, 5]; the Johnston–Yang and Maynard records likewise remain inherited source-closure objects only. No new conclusion in this paper is attributed directly to those four remote records.

8.2 Finite reproduction and source integrity

The companion certificate is an exact finite reproduction artifact, not an analytic proof. Its frozen canonical JSON has 96 rows, 83,309 bytes, and SHA-256 `7cc0da78ee7e47a22b357d7e8d907bc9d9879caeb82ede30709e8cb1023032ba`. The strict validator rejects all 32 named single-core mutations. Among its checks, the exhaustive full-eight relation oracle scans 262,144 ordered relation pairs and finds exactly 3,375 universally safe pairs. These checks reproduce finite identities and counterexamples used above; they do not replace the proofs.

The frozen source closure contains 160 immutable Git objects and four inherited remote records, hence 164 logical objects. Its canonical SHA-256 is `c16456d58efd74edf1505c430a54459e359b5ba7e1e581773e9a0613b493385b`; the all-Git and logical-source digests are, respectively, `472bf5ce5e352dce0d3a44ad10b22345b98e0e8b9a0cd745be9ecd93dedf0a86` and `72040ab3d7a5d98ce308b91d0748d52a8d4886cf245f5079f14c69ee659cc287`. Reproduction is offline: the release records zero network requests, vendors no external PDF payload, and records no forbidden payload-hash hit. The four remote redistribution flags are (no, no, yes, no); the corresponding PDFs are not redistributed.

8.3 Claim firewall

Every analytic assertion fixes h , q , and the complete phase table before $X \rightarrow \infty$, and it holds for every terminal clock in (1). The order is: terminal limit for each fixed table, finite safe maximum, and only then the scalar supremum over finite clocks. The result is not an ordinary Cesàro-average theorem and contains no regime with $h = h(X)$, $q = q(X)$, or growing phase tables; no uniform rate in these parameters is asserted.

The centered rule reads $\mu(n + h)$, so it is not causal or online. The proof concerns exactly three shifts and safety at the single separation $2h$; it does not establish an even four-shift law, a larger-window compiler, or a generic graph theorem. Nothing here is an analytic trace or operator

formula, a statement about zeta zeros or the Riemann hypothesis, or a discharge of downstream Gates A–E. Finally, the lag argument proves only the infimum and its nonattainment. It makes no assertion about a supremum, a maximum, or monotonicity in h .

Declarations

Data availability. The finite certificate, exact result and schema, source-lock metadata, and offline verification code are included in the accompanying RH-396 package. No new empirical data were generated and no external source PDF is redistributed.

Ethics declaration. This mathematical study involved no human participants, animals, personal data, or sensitive datasets; institutional ethics approval was not applicable.

Author contributions (CRediT). The RH research program performed conceptualization, formal analysis, methodology, software, validation, and writing (original draft, review, and editing).

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AI-use statement. AI-assisted tools supported drafting, finite-certificate implementation, formatting, and consistency checks. The research program directed and reviewed the mathematical arguments, source roles, and release artifacts and retains responsibility for the content.

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